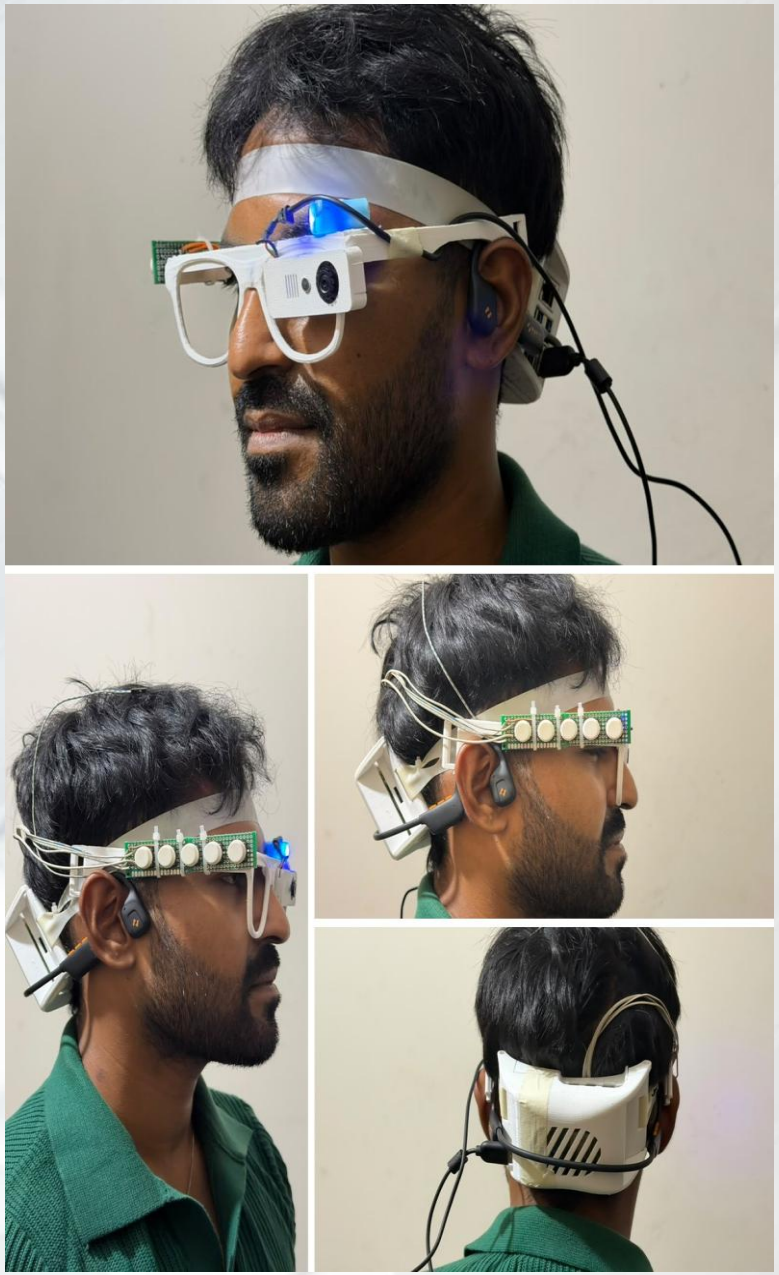
SARA · Smart Assistive Reading Aid

A Wearable Assistive Glass for Authenticating Bangladeshi Banknotes: Fusing Physical Security Features and Deep Learning for Visually Impaired Users

A bilingual (Bangla and English) smart glass that also reads printed Bangla and English text and detects everyday objects, offline on a Raspberry Pi 5

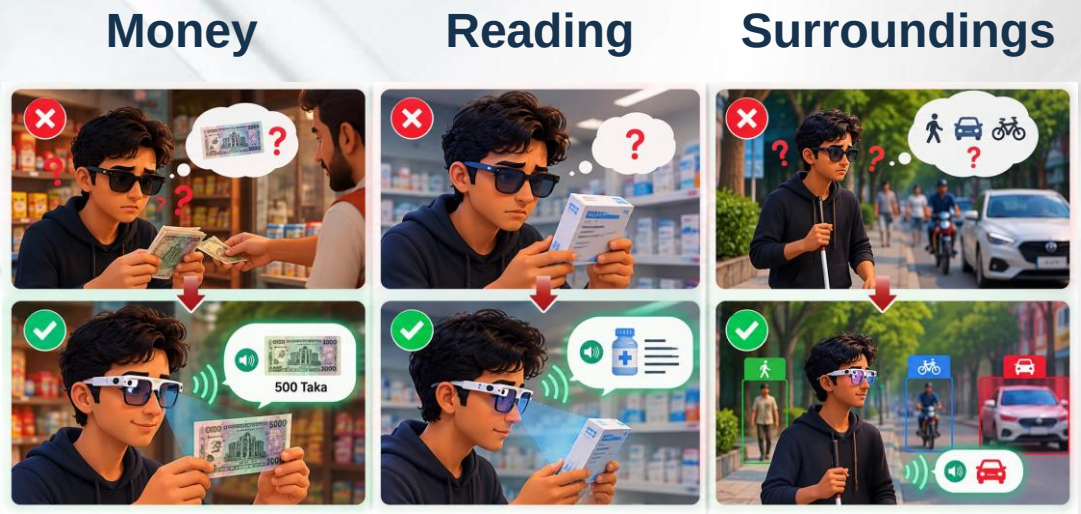


Supervisor:
Mohammad Shidujaman, PhD
Assistant Professor
Department of Computer Science & Engineering

Submitted By:
MD Abdun Fattah Lam (2220014)
Mukut Protim Memo (2020086)
Tanjila Hossain Tapti (2130561)

Introduction

- At least 2.2 billion people have a near or distance vision impairment, and its prevalence in low- and middle-income regions is about four times that in high-income regions [1]
- In Bangladesh, three everyday questions usually require help from another person:
  - **Money:** which note was handed back, and is it genuine?
  - **Reading:** what does a medicine label, sign or letter say?
  - **Surroundings:** what or who is ahead?
- In each case the blind person depends on an answer they cannot verify
- **SARA is an offline, bilingual assistive glass built for this setting**



Top row: today, an answer from another person. Bottom row: with SARA, the glass answers. (Illustration)

Problem Statement

Visually impaired user



receives change → **Money: at the shop**



must trust → **Shopkeeper**

needs a label read → **Reading: at the pharmacy**



asks → **Family or pharmacist**

walks outside → **Surroundings: on the street**



asks the way → **A stranger**

gives an answer

An answer they cannot verify

- A wrong note or a fake note can go unnoticed
- A label or letter can be misread
- Every task waits for someone else

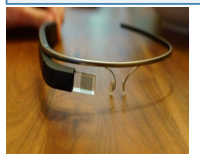
Existing aids do not close this gap

OrCam MyEye [2]



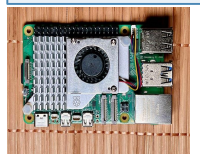
Offline, but no Bangla and no Taka

Research glasses [3, 4]



Pictures go to a server

Pi reading aids [5]



Mostly English-only

Counterfeit studies [6, 7, 8, 9, 10]



Not on a wearable

The gap
No reviewed device is offline, speaks Bangla and names Taka notes

Proposed Solution: SARA

User wearing SARA (staged)



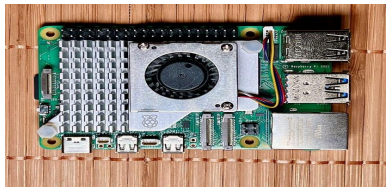
button, or "Glass, check money" [11]

Glass camera



captures a picture

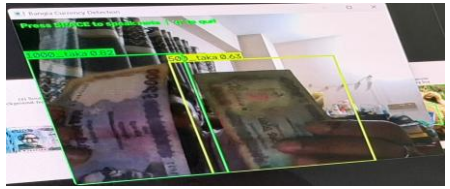
Raspberry Pi 5 [12]



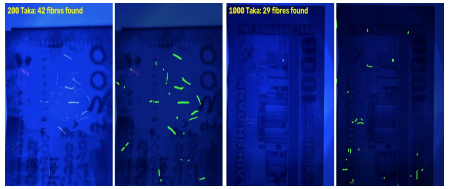
Offline: text, objects, Taka, emotion
Online (optional): scene description only

processes offline

Taka recognition [13]



Authenticity: watermark + UV



Text reading [14, 15]



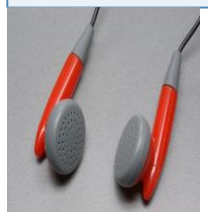
Objects and people [13, 16]



Safe verdict (money)

"Likely genuine" or "check by hand whether it is genuine"; it never says "counterfeit"

Spoken answer in Bangla or English [17] + a short vibration

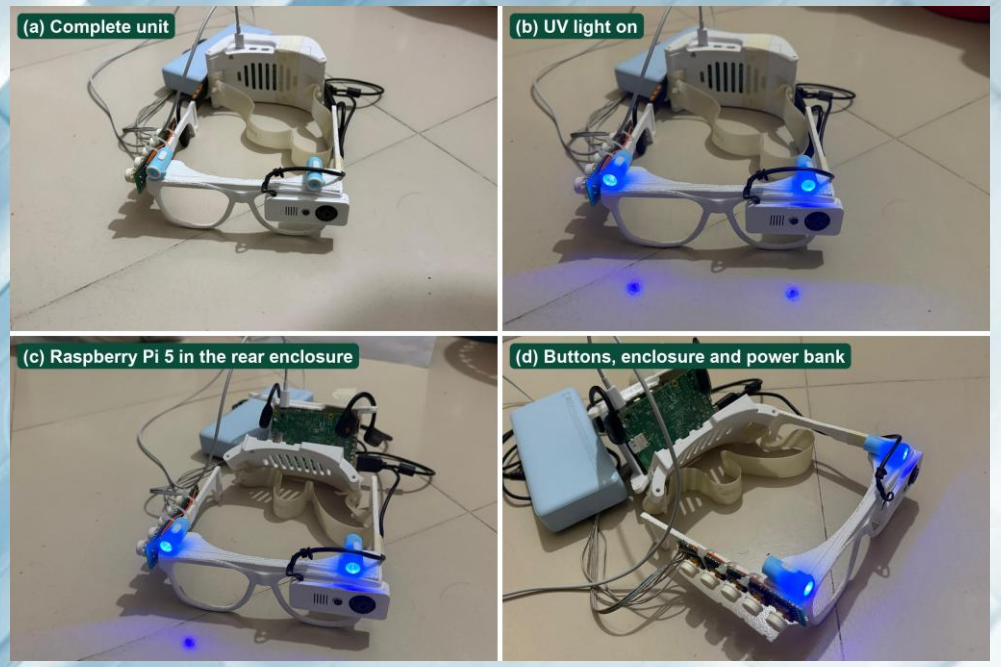




পাঁচশত টাকার নোট

"500 Taka note": known without asking anyone

System Features



Prototype version 2: the complete unit, the UV light on, the Pi 5 in the rear enclosure, buttons and power bank

Text reading

Bangla and English

Objects

Named in Bangla

Taka notes

Value + safe check

Emotion

Face in front

Online description

Optional; needs internet

How the user starts a mode: five buttons on the frame, or a voice command: "Glass, read text"

Models: text [14, 15] CRAFT + EasyOCR; objects [13, 16] YOLOv8 on COCO; emotion [18] RAF-DB; voice commands [11] Vosk.

Research Questions

1 System

Can it all run on a Raspberry Pi 5?

2 Synthetic data

Can unlabelled training data detect real notes?

3 Number of views

Can one view of a note be enough?

4 Unseen prints

Does the check hold on new counterfeit prints?

5 Safety

Can it never accuse a genuine note?

6 Bangla OCR

What matters most for reading Bangla?

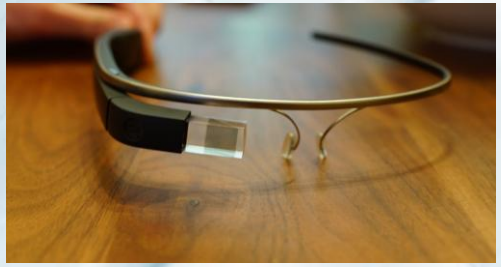
Literature Review: Existing Assistive Devices

System	Reads Bangla	Knows Taka	Works offline	Wearable
OrCam MyEye [2]	No	No	Yes	Yes
Google Glass scene analysis [3]	No	No	No	Yes
AIris (2024) [4]	No	No	No	Yes
Raspberry Pi reading aids *	No	No	Partly	Partly
SARA (ours)	**Yes**	**Yes**	**Yes**	**Yes**

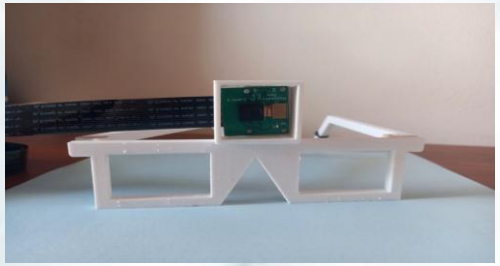
*Rows with a reference: from the cited source. * Our assessment of typical hobby builds, not a cited result.*



OrCam MyEye [2]



Google Glass analysis [3]



AIris [4]

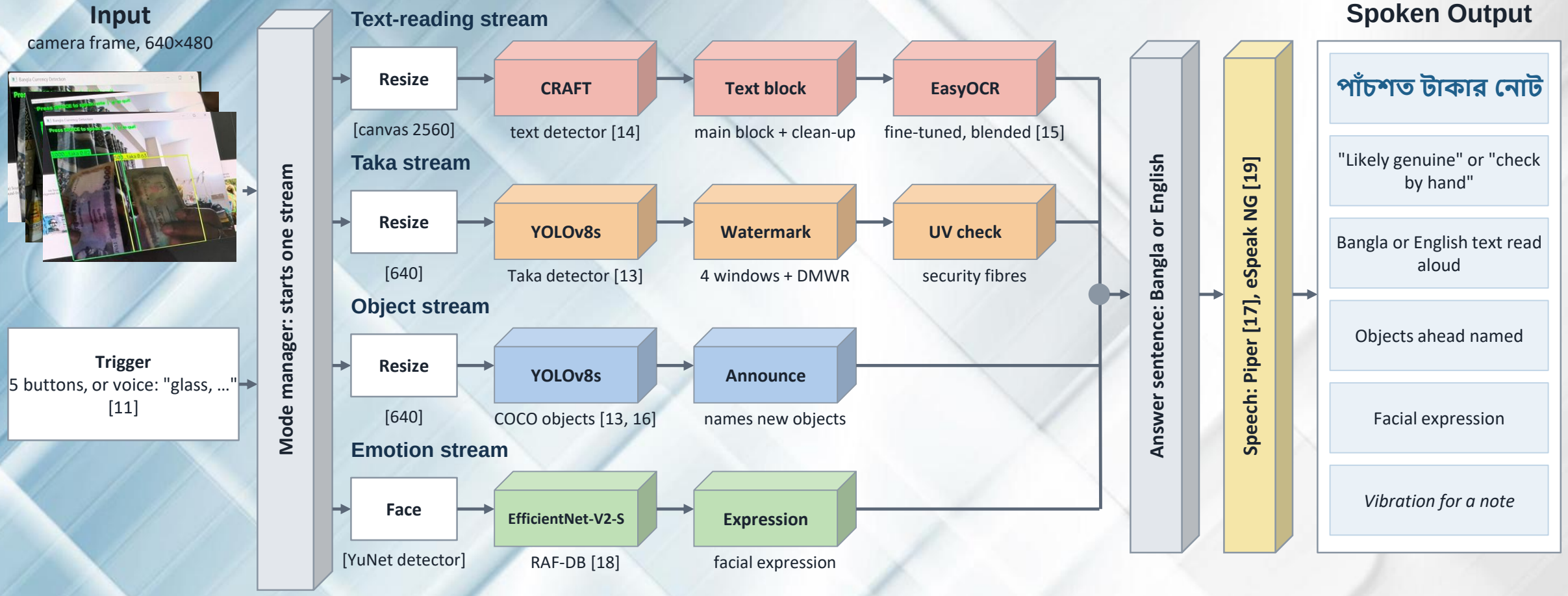
The research gap

None of the systems we reviewed combines Bangla, Taka and offline use.

- Counterfeit tests reuse the same prints
- Few report speed on the real device

Photos: OrCam MyEye by Ylip, CC BY-SA 4.0; Google Glass by Ted Eytan, CC BY-SA 2.0 (Wikimedia Commons); AIris from [4]. Comparison from [2, 3, 4].

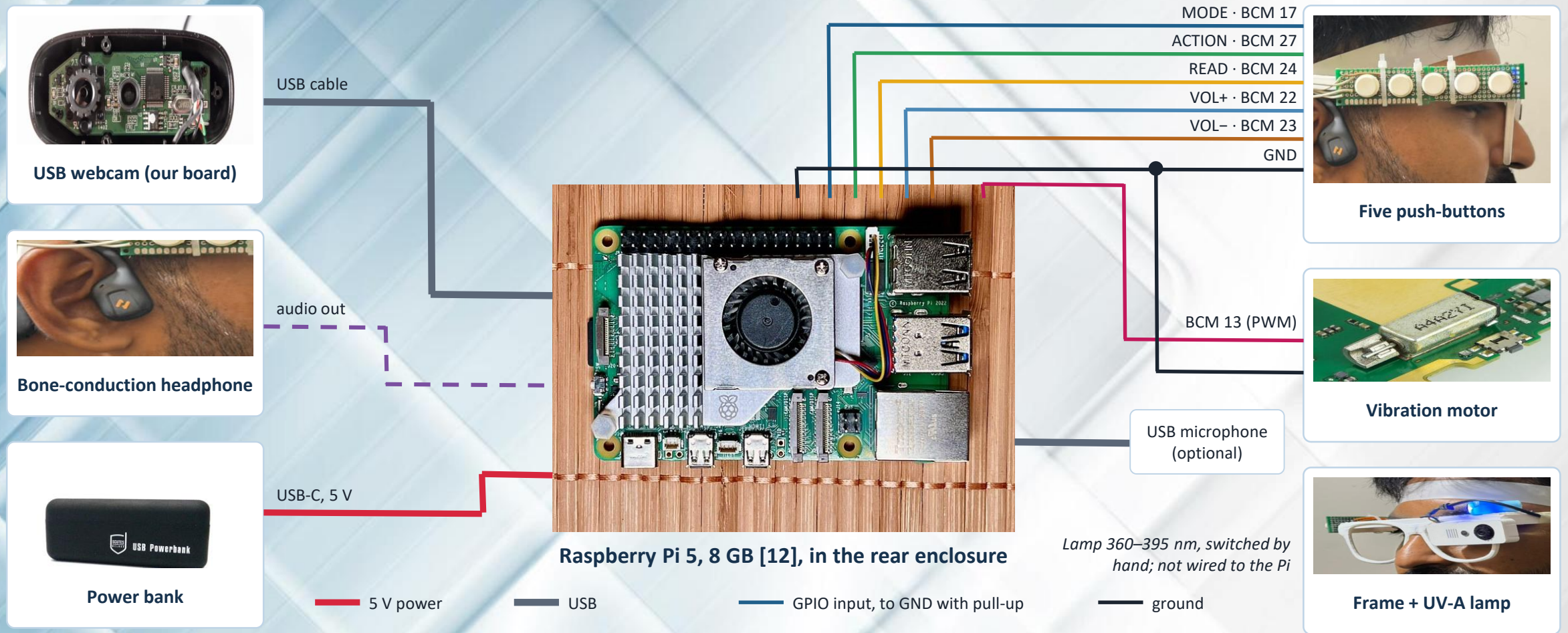
System Architecture



All streams run on the Raspberry Pi 5 CPU, offline. The optional online scene-description mode is not shown.

Sources: [11] Vosk; [13] Ultralytics YOLOv8; [14] Baek et al. 2019; [15] EasyOCR; [16] COCO; [17] Piper; [18] RAF-DB; [19] eSpeak NG.

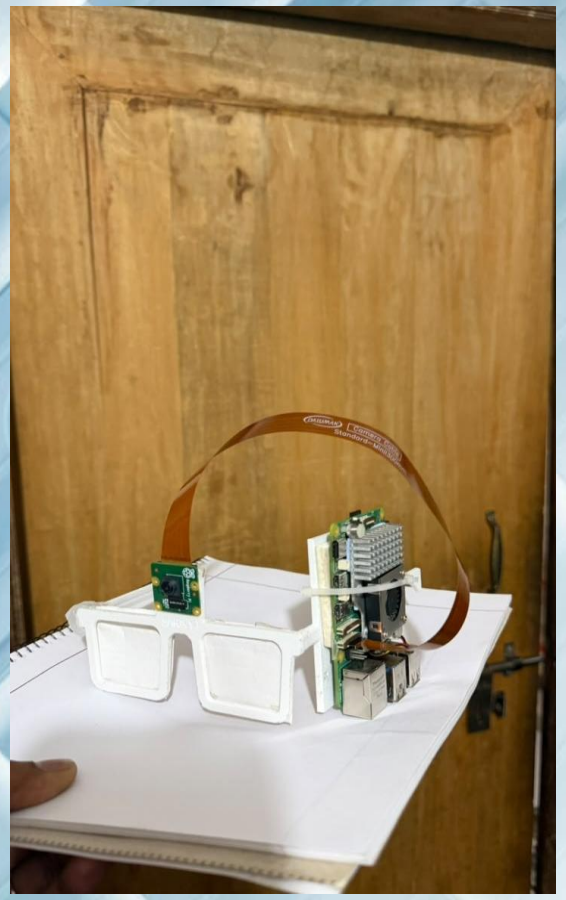
Hardware Design



Schematic, not to scale; pin numbers are those set in the code.

Pictures (Wikimedia Commons): Raspberry Pi 5 by SimonWaldherr, CC BY 4.0; vibration motor by Raimond Spekking, CC BY-SA 4.0; power bank by Pinguinguy, CC0. Webcam board, buttons, headphone and frame: our prototype.

Evolution of the Prototype



Version 1
Board on one temple



Front
Camera + UV lamp



Side
5 buttons, bone-conduction earphone



Rear
Pi 5 in a ventilated case

Second Prototype: Its Parts

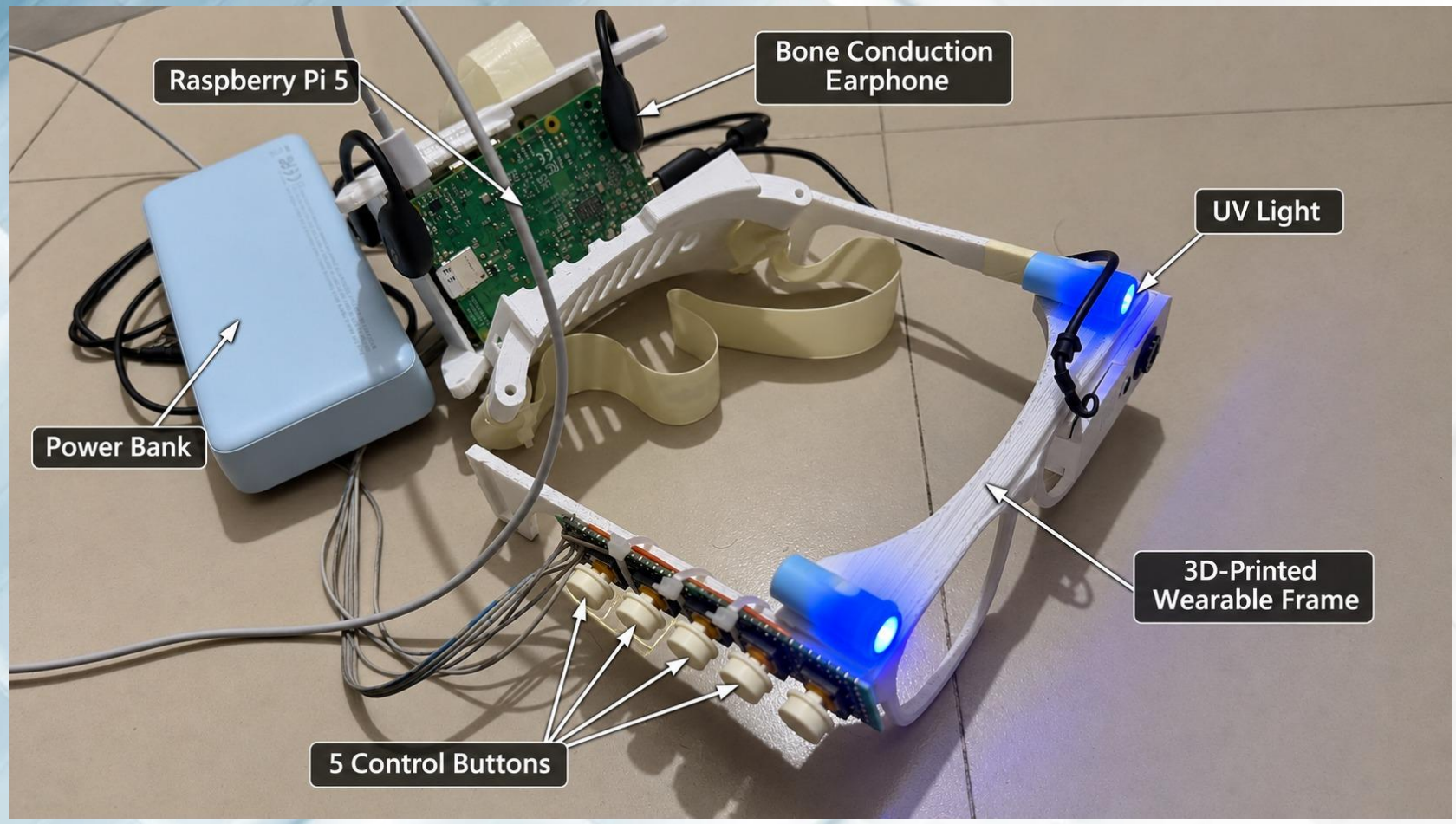


Photo: our second prototype (team photo, labels added by the team).

Banknote Detection: Synthetic Training Data



Note photos [20] are pasted onto real scenes [16], and the program writes every label itself.

SYNTHETIC COMPOSITES

0.995

mAP@0.5 on 2,282 held-out composites

REAL PHOTOS

94.1%

right note value on 1,536 photos taken by other people (laptop)

Sources: [20] BanglaTaka (Mendeley); [16] Lin et al., COCO, ECCV 2014; detector [13] Ultralytics YOLOv8. 94.1% with the hand-held detector (next slide).

Banknote Detection: Hand-Held Notes

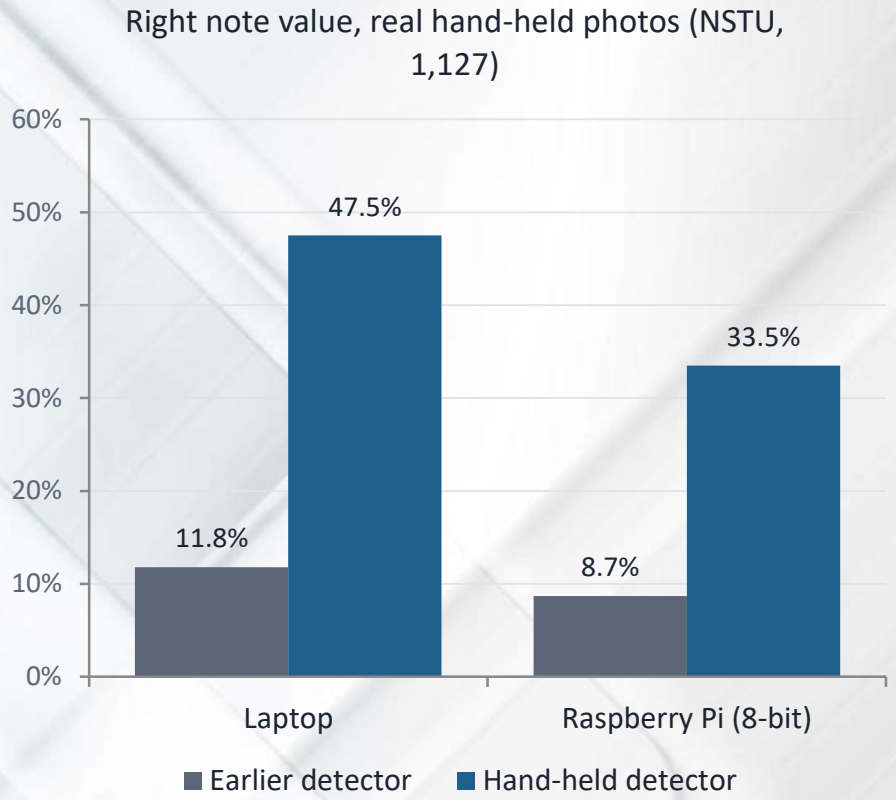
TRAINING: 6,000 SYNTHETIC CLOSE-UPS



TEST: REAL HAND-HELD PHOTOS

11.8% → 47.5%

The glass runs the same full-precision detector as the laptop. Trade-off: with 3 notes in view, the total is right 97.0% (was 99.2%).

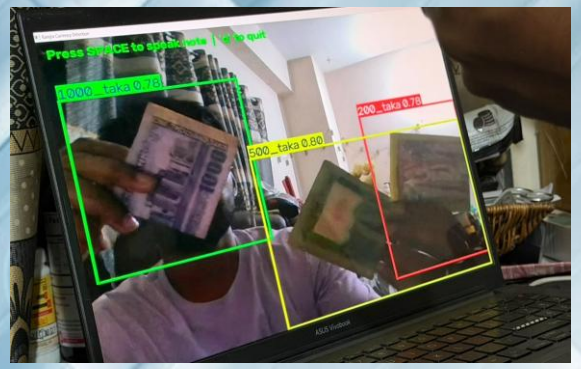


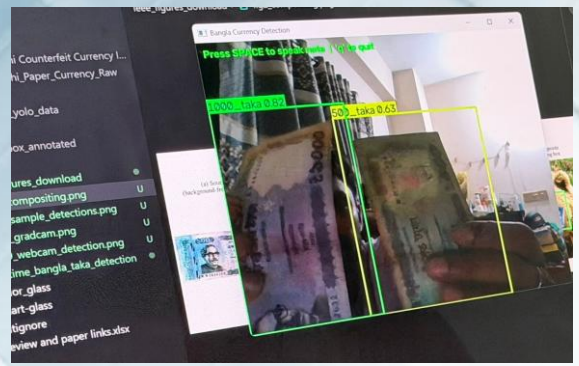
Sources: test photos [21] Rafi et al., NSTU-BDTAKA, Data in Brief 2024; detector [13] YOLOv8. Our results: paper_evidence/HANDHELD_FINETUNE.md.

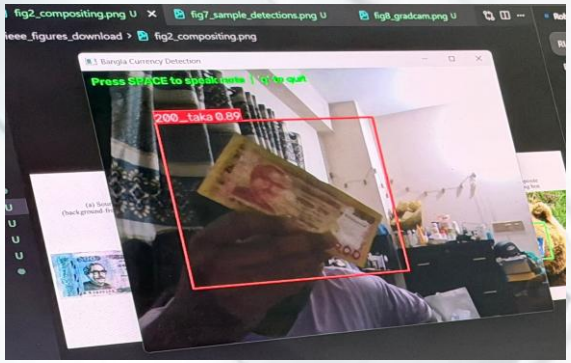
Live Detection Tests

Informal demonstrations, not a measured accuracy study

Software: live detection on a laptop webcam (detector output shown)









Second prototype (v2): live tests with the glass (staged)



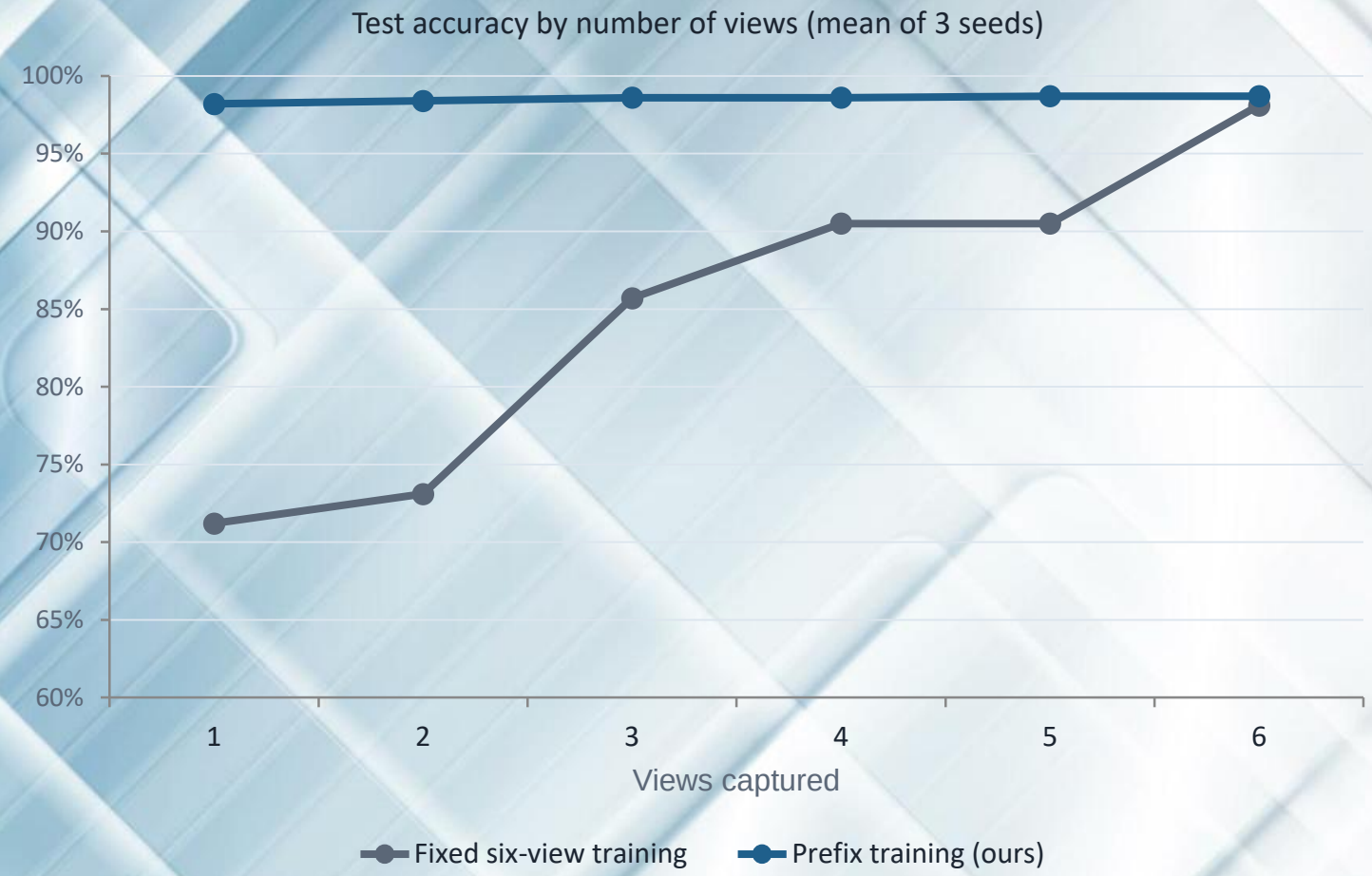






Screenshots: 50, 200, 500 and 1,000 Taka named at 0.63–0.89 confidence, two or three notes at once. The v2 photos do not show the detector output.

Counterfeit Screening: Prefix-Robust Multi-View Training



With a single view

71.2%

rises to

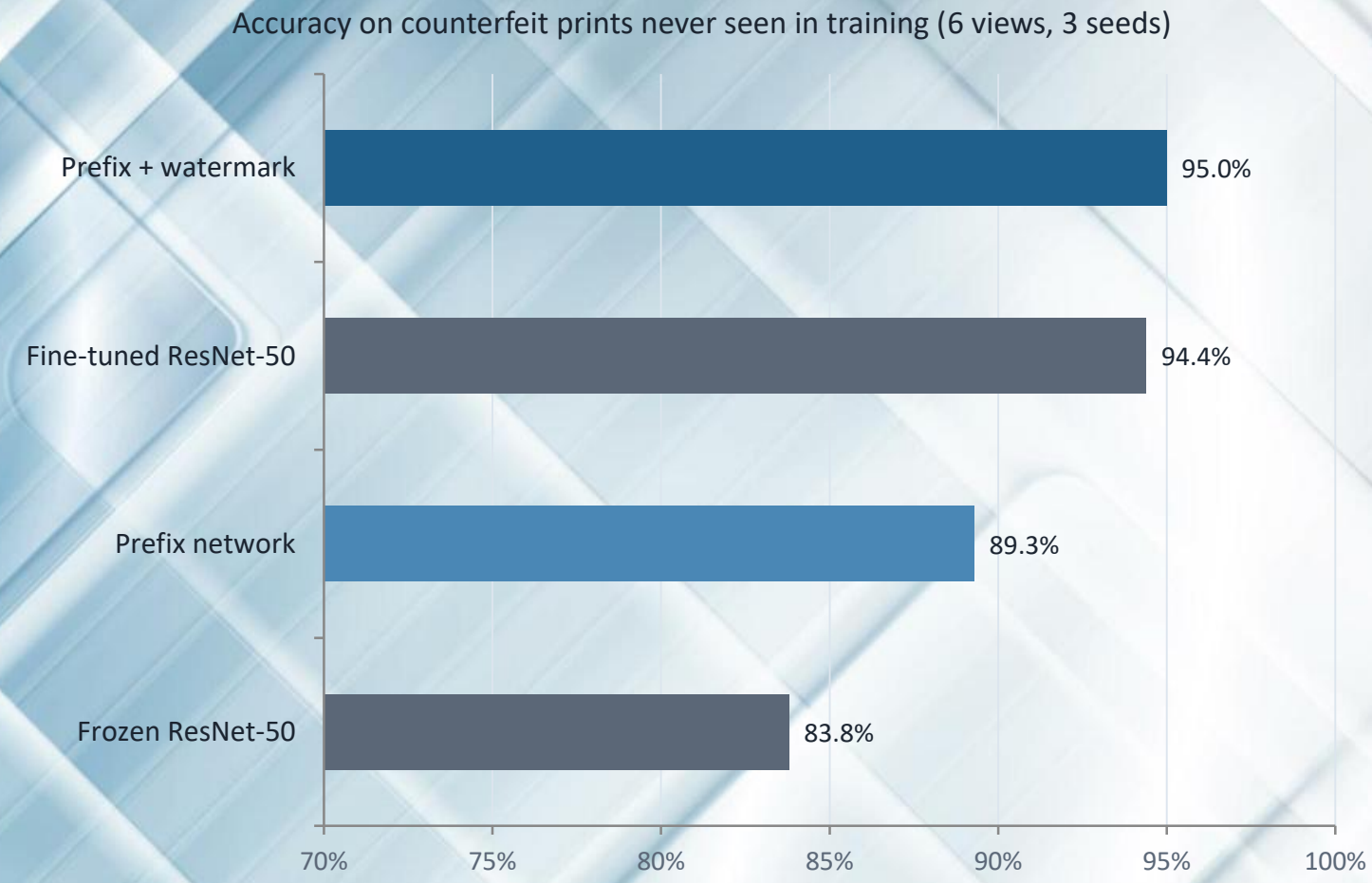
98.2%

Training on random first views: one view is enough.

Held-out JaalTaka test notes, never used for training or for any choice

Data: [9, 10] Piyas et al., JaalTaka dataset and article, 2025-2026 (1,390 notes, six views each; counterfeits seized by the Rapid Action Battalion).

Counterfeit Screening: Evaluation on Unseen Prints



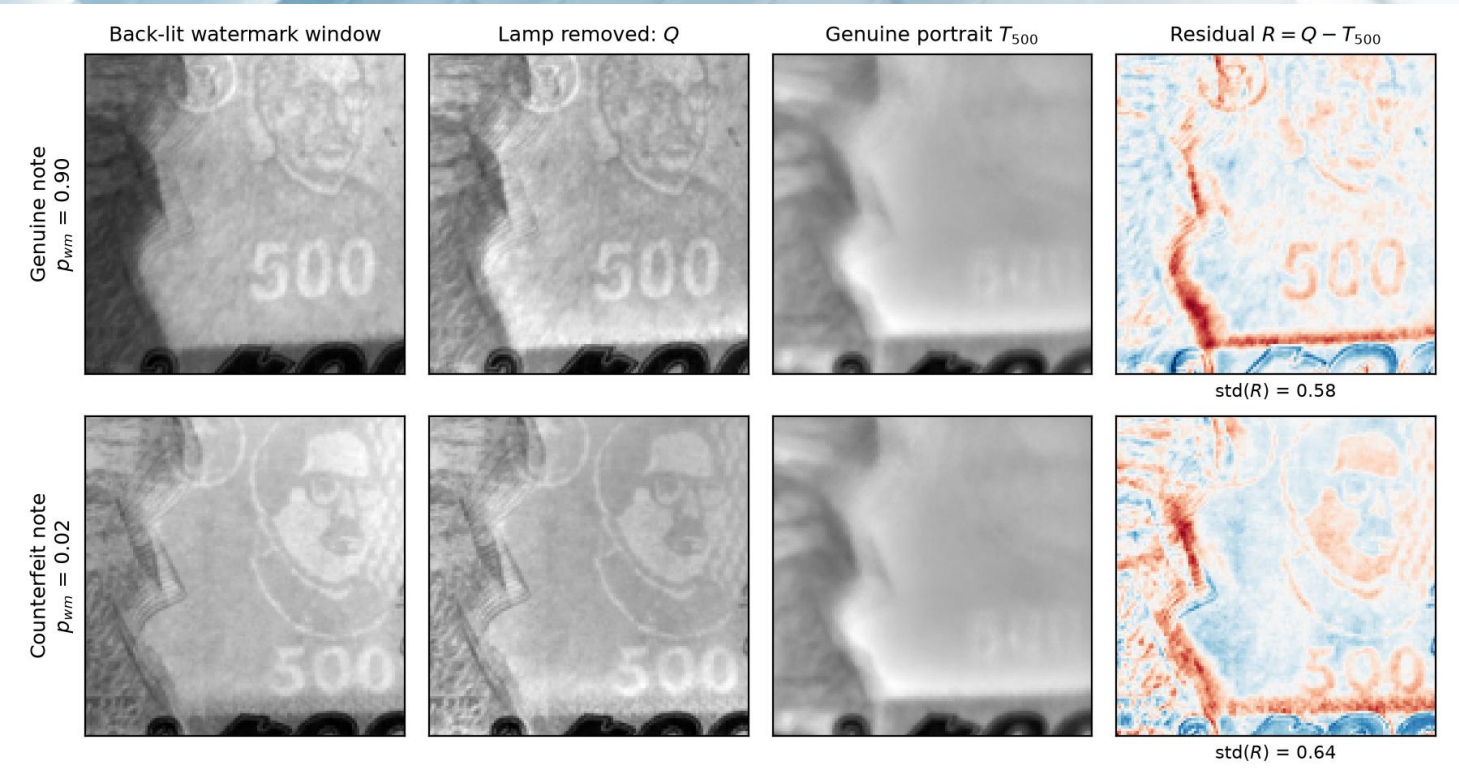
95.0%

on counterfeit prints never seen in training, with the watermark check

Why a stricter split: 279 of 322 readable 500 Taka counterfeits share one serial number, so the usual split tests copies of known prints.

Data: [9] JaalTaka; serial-number audit and print-disjoint split are ours. Earlier Taka counterfeit work: [6] SVM on security features; [7] CNNs; [8] UV flecks + RPN.

Watermark Verification: DMWR



1. Remove the lamp: Q = I / (G∗I)
2. Genuine portrait: T = median of Q over genuine notes
3. Residual R = Q − T → two small CNN streams → "watermark clear?"

91.5%

on unseen counterfeit prints (3 seeds, ±0.3)

46× smaller

than MobileNetV2 [22] (47,818 parameters, 0.21 MB), and as accurate: 92.9%, p ≥ 0.375

Baseline [22] Sandler et al., MobileNetV2, CVPR 2018; data [9] JaalTaka. DMWR (denomination-matched watermark residual) is our method.

Safe Verdict Policy



0 / 88

test counterfeits passed as "likely genuine"

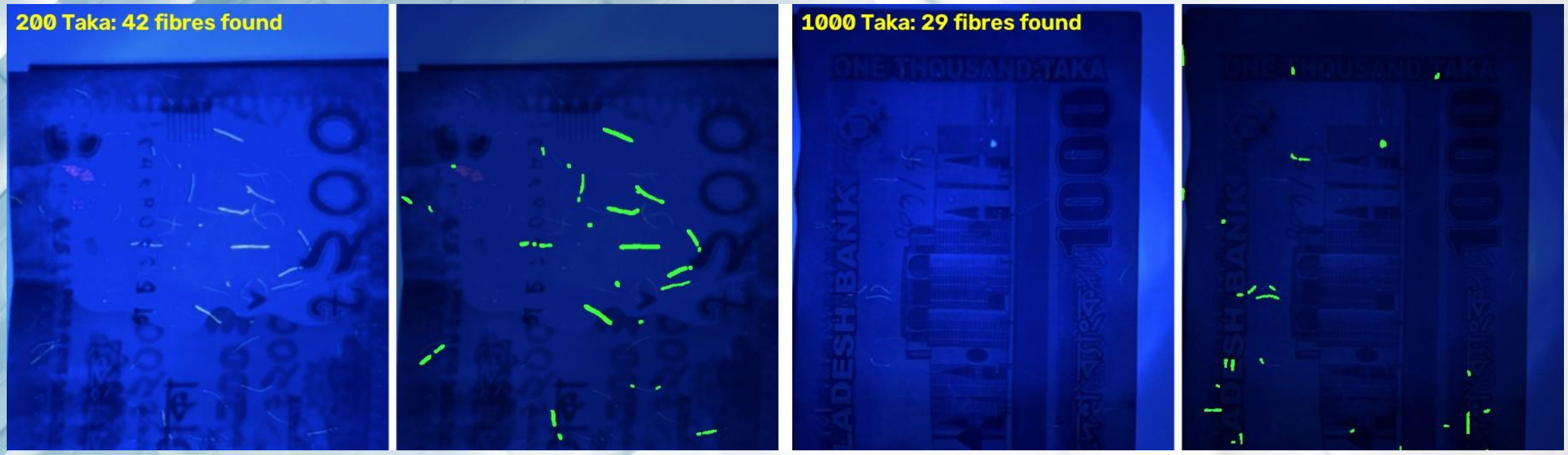
"Likely genuine": the score is higher than every counterfeit seen in validation (57.5% of genuine notes).

"Check by hand": the glass is not sure. It never means "counterfeit".

A screening aid, not a guaranteed counterfeit detector

Test notes: [9] JaalTaka.

UV Security-Fibre Verification



85.6%

of 111 real UV photos of genuine notes accepted

88.6%

of SIMULATED fibre-free sheets rejected (not real counterfeits)

Speaks under the lamp

"Likely genuine" or "check by hand", never "counterfeit". Not yet tested on real counterfeits or through the glass camera

The glass's lamp emits 360-395 nm, covering the 365 nm used for banknote checks; banknote paper stays dark while its fibres glow [23]. Prior Taka UV work: [8] (96.3%). Fibre check and photos are ours.

Object Detection

How it works

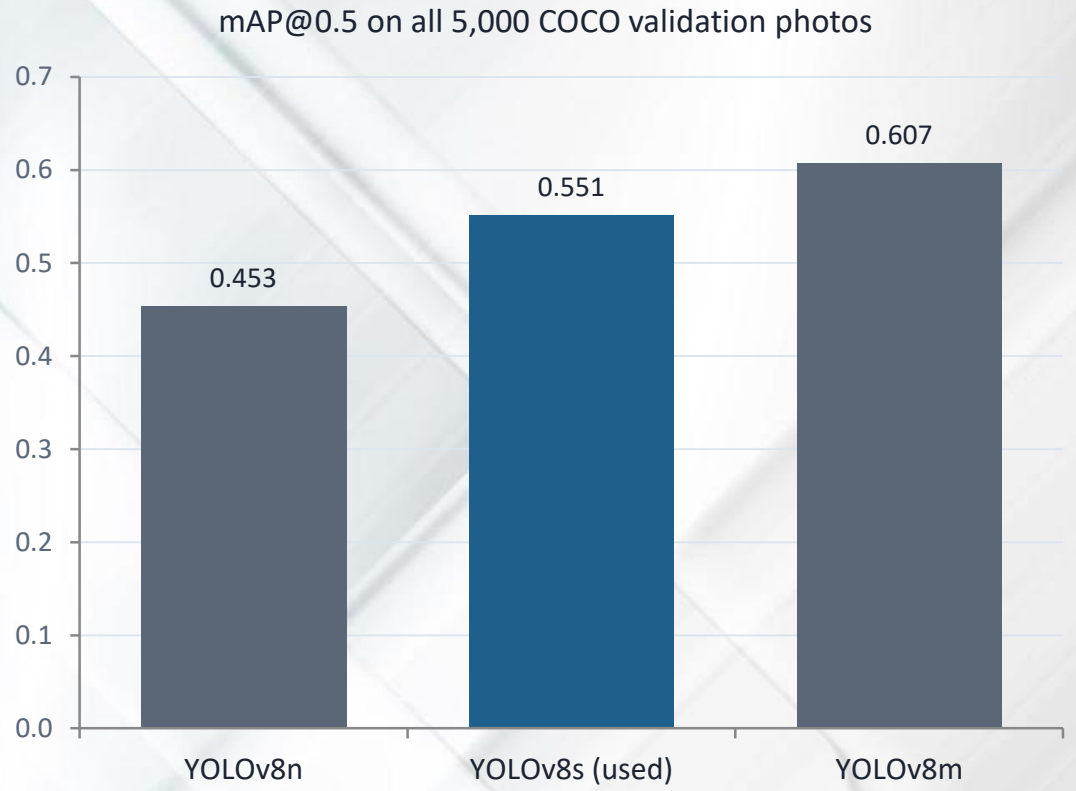
- Every 1.5 s a camera frame goes to YOLOv8s [13, 16]: 80 everyday object types
- Only new objects are spoken; the same object is not repeated within 4 s
- ACTION announces at once
- Animals need 65% confidence: in dim rooms, clutter was called "cat" or "dog"

সামনে আছে: মানুষ, মোবাইল ফোন

spoken by the glass in a test session

462 ms

one scan on the Raspberry Pi 5



YOLOv8s balances accuracy and speed on the Pi. Through the glass's own path it scores 0.521.

Detector [13] Ultralytics YOLOv8, trained on [16] COCO. Results: savior_glass/results/object_coco_val2017_full.json, object_shrink_vs_full.json; Pi time from benchmark_report/.

Facial Expression Recognition

How it works

- Finds the largest face in the picture
- EfficientNet-V2-S trained on RAF-DB [18] (12,271 faces), blended 3 : 1 with FER+
- Speaks one of 7 expressions in Bangla

সামনের মানুষটিকে খুশি দেখাচ্ছে

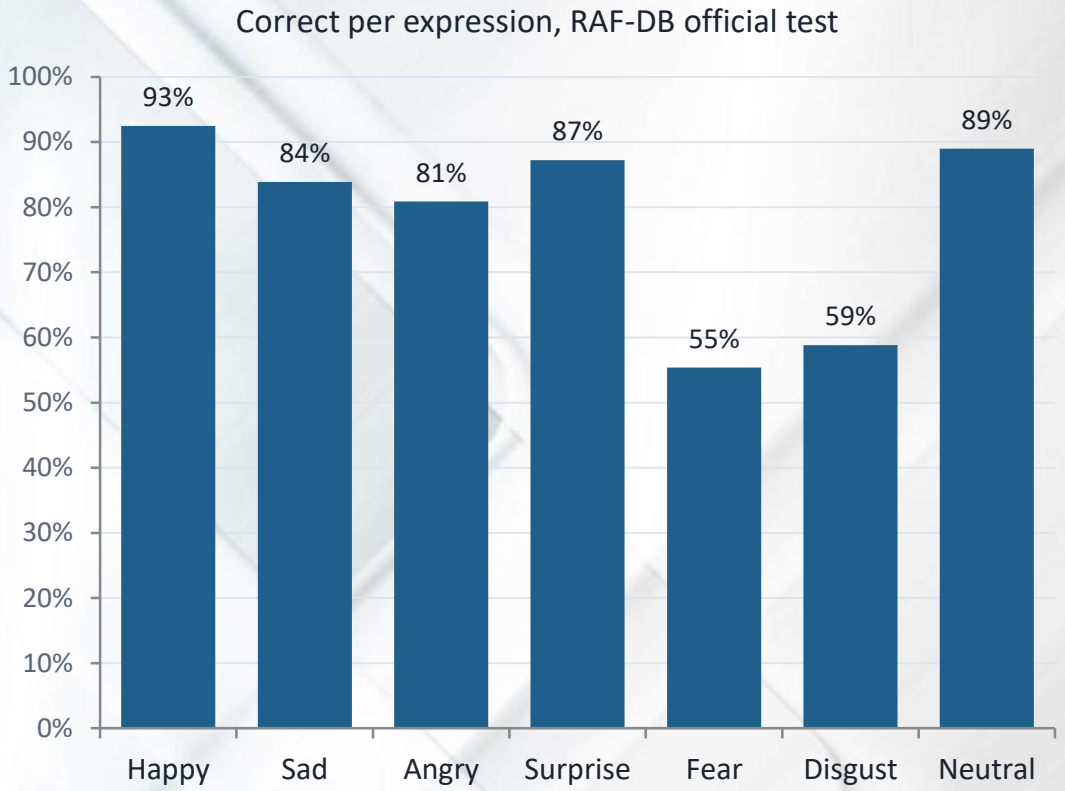
"The person in front looks happy"

86.5%

correct on the RAF-DB test (3,068 faces)

536 ms

one answer on the Pi 5, face in view

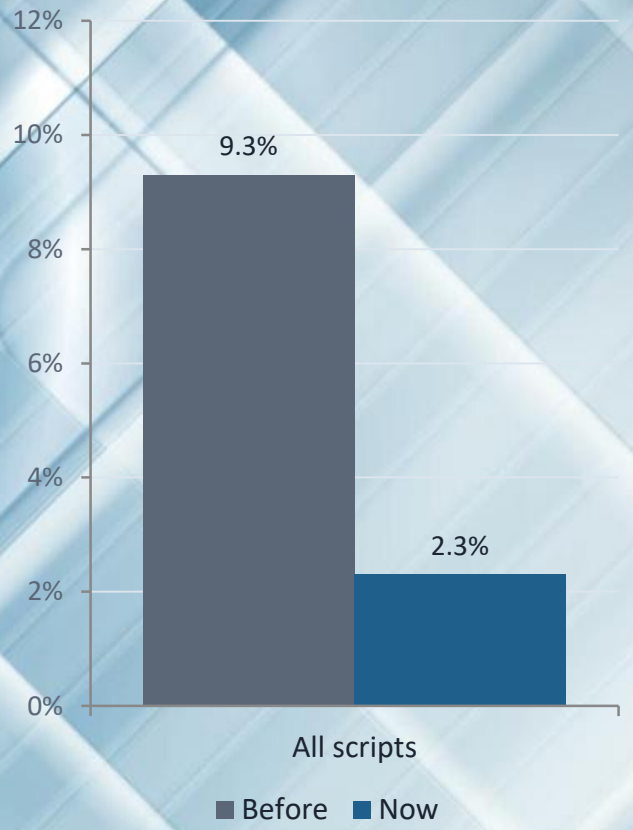


Fear and disgust are hardest (fewest examples). Measured for the network; the blend on the glass was not tested separately.

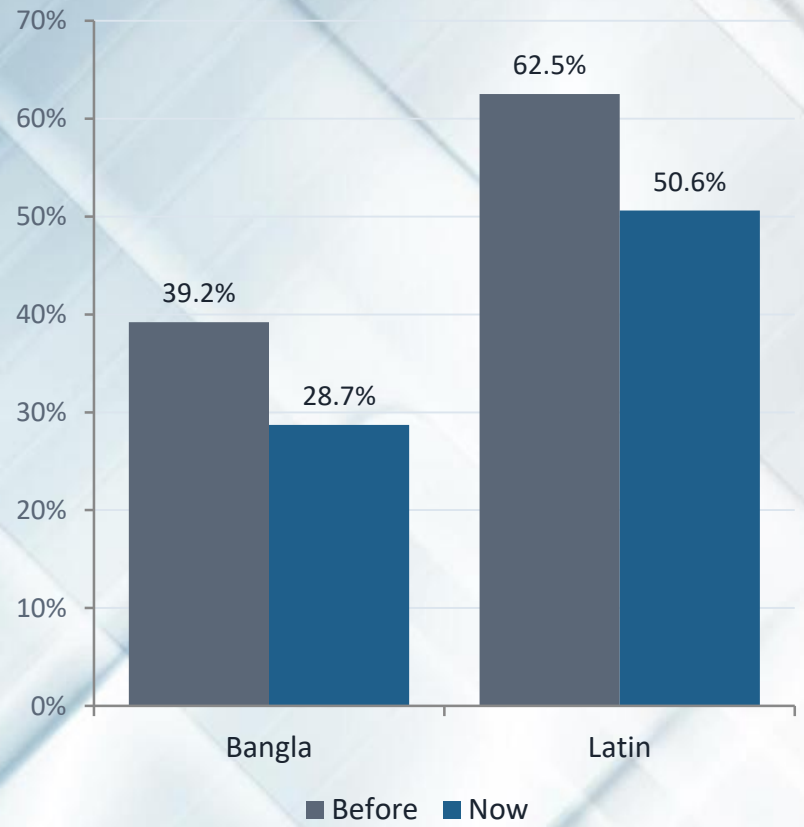
Data [18] RAF-DB, official train / test split. Results: savior_glass/results/emotion_rafdb.json; Pi time from benchmark_report/.

Bangla and English Text Recognition

Rendered text (clean)



Real word photos (hard)



What helped

- Reading the main text block only
- Tuned detector and clean-up
- Recogniser fine-tuned on real words
- Sharpest frame of 1 s: shaken text 28.0% → 3.8%

What did not

Binarisation, Tesseract, TrOCR, PaddleOCR, dictionaries

Character error rate, lower is better. Measured on evaluation datasets, not yet through the glass.

Engine [15] EasyOCR with [14] CRAFT; real words from [24] ICDAR 2019 MLT (1,000 test words). Every choice made on validation data.

Text-to-Speech: Spoken Answers

How the glass speaks

- Every answer becomes one Bangla or English sentence; the language is found per sentence
- Piper neural voices [17], offline; eSpeak NG [19] as a fallback
- Each sentence is made in full, then played; the next one is prepared meanwhile, so the sound does not break when the Pi is busy
- Sentences stay whole (they used to be cut at every comma and digit)

14 ms

to make a short Bangla sentence on the Pi (eSpeak NG)

0.2–0.5 s

for a Piper Bangla sentence (laptop)

Is the Bangla voice understood?

The glass said	The listener heard
পাঁচশত টাকার নোট সম্ভবত আসল	পাঁচশত টাকার নোট সম্ভবত আসল
ধূমপান নিষেধ	ধূমপান নিষেধ
এক হাজার টাকার নোট	1000 টাকা নোট
কোনো লেখা পাওয়া যায়নি	কোন লেখা পাওয়া যায়নি
সামনে আছে চেয়ার টেবিল	স্বামী আছে যে টেবিল
নোটটি আলোর সামনে ধরুন	নটি আলু সামনে ধরুন

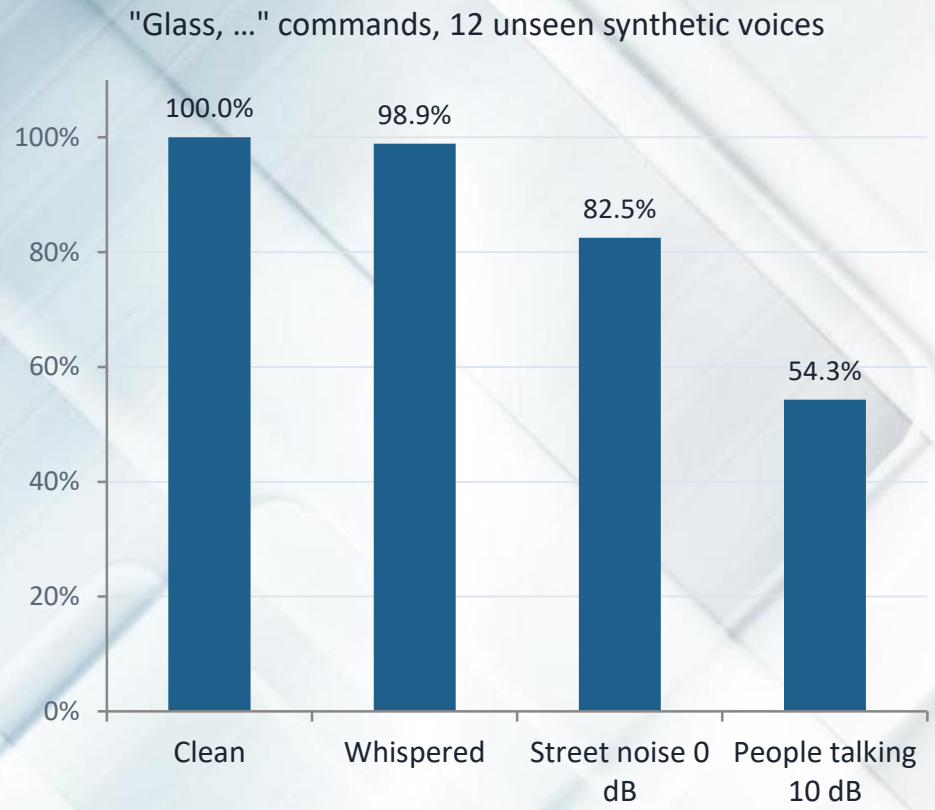
2 exact, 2 small differences, 2 with wrong words. A machine listener, not people.

Voices [17, 19]. Results: savior_glass/results/speech_piper_bn_intelligibility.json (Google Web Speech, bn-BD, as the listener); Pi time from benchmark_report/.

Voice Interaction

Speaking

- Natural Piper voice, Bangla and English [17, 19]
- No broken sound when the Pi is busy
- "পড়ছি" (reading) while text is captured



212 → 13

false commands in 5,040 ordinary sentences, thanks to the wake word

0.86 s

from the end of speech to the action (median)

Command results: synthetic voices on a laptop, not human trials on the glasses

Recogniser [11] Vosk; noise from [25] Piczak, ESC-50; voices [17, 19]. Results: paper_evidence/VOICE_COMMANDS_FAST.md.

Performance on the Raspberry Pi 5

532 ms
currency check, end to end

50 ms
watermark check

462 ms
object scan (YOLOv8s)

536 ms
emotion, face in view

5.4 s
text reading, 3-line sign

2.76 GB
peak memory of the app (of 8 GB)



Board temperature

57.8–62.3 °C

measured on the board with a thermometer while the glass was running

The processor's own sensor reads higher: in 30 minutes of non-stop load it peaked at 87.0 °C, with the soft limit active in 55% of samples.

Medians measured on the glass's own Raspberry Pi 5 (8 GB). With the glass in use, a note was named in 5 of 8 presses, 0.81 s from press to speech.

Board: [12] Raspberry Pi 5 (8 GB), fan and heat sink, C270 webcam. All timings are ours (benchmark_report/); board-surface temperature measured by the team with a thermometer.

Prototype Testing and Project Status



Prototype version 2 in a staged shop exchange by team members; illustration only, no error rate counted.

DONE **Engineering checks**

Each part tested on held-out data, on a laptop and the Pi 5

PRELIMINARY **First field test**

Glass camera: currency unreliable, Bangla text not read

IN PROGRESS **Participant trials**

Sessions with a visually impaired participant have started

NONE YET **Full user study**

No results about users are claimed

Future Directions



Camera

Higher resolution or autofocus

User study

8–12 participants, including their own voice commands

More banknotes

Unseen counterfeit prints; UV check on real fakes

Speed and cooling

Time the new code on the Pi; better cooling; battery

Done: engineering tests **In progress:** participant trials **Future:** the four steps above

Conclusion



Built

An offline Raspberry Pi 5 glass that reads Bangla and English, names objects and Taka notes, and speaks Bangla

Investigated

Synthetic training data, prefix training, unseen prints, a safe verdict, DMWR, hand-held notes, OCR, voice

Still to validate

Real counterfeits (also under UV), people's voices, the glass camera, a completed user study

THANK YOU

Questions?

References

[1] World Health Organization. Blindness and vision impairment: fact sheet, updated 10 February 2026.

[2] OrCam Technologies. OrCam MyEye.

[3] Ali A., H. et al. A Google Glass based real-time scene analysis for the visually impaired. IEEE Access 9, 2021.

[4] Brilli, D. D. et al. AIris: an AI-powered wearable assistive device for the visually impaired. arXiv:2405.07606, 2024.

[5] Adusumilli, V. et al. Reading aid and translator with Raspberry Pi for blind people. ICACCS, 2023, pp. 327-331.

[6] Uddin, M. S., Das, P. P., Roney, M. S. A. Image-based approach for the detection of counterfeit banknotes of Bangladesh. ICIEV, 2016.

[7] Pathak, A. et al. Enhanced counterfeit detection of Bangladesh currency through convolutional neural networks. IJIRCST 12(2), 2024.

[8] Chowdhury, A. A. et al. Evaluation of deep learning models on UV ink: a fake money detection scheme with RPN. ICCA (ACM), 2022.

[9] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Mendeley Data V2, 2025.

[10] Piyas, B. R. C. et al. JaalTaka: a benchmark dataset for detecting counterfeit Bangladeshi banknotes. Data in Brief 64, 112377, 2026.

[11] Alpha Cephei. Vosk offline speech recognition toolkit.

[12] Raspberry Pi Ltd. Raspberry Pi 5 product brief, 2023.

[13] Jocher, G., Qiu, J., Chaurasia, A. Ultralytics YOLO, version 8.0.0, 2023.

[14] Baek, Y. et al. Character region awareness for text detection (CRAFT). CVPR, 2019.

[15] JaidedAI. EasyOCR: ready-to-use OCR with 80+ languages.

[16] Lin, T.-Y. et al. Microsoft COCO: common objects in context. ECCV, 2014.

[17] Rhasspy. Piper: a fast, local neural text-to-speech system.

[18] Li, S., Deng, W., Du, J. Reliable crowdsourcing and deep locality-preserving learning for expression recognition in the wild (RAF-DB). CVPR, 2017.

[19] eSpeak NG text-to-speech.

[20] Nuhash, M. N. I., Akter, S., Mojumdar, M. U. A diverse image dataset for Bangladeshi currency recognition (BanglaTaka). Mendeley Data V1, 2024.

[21] Rafi, M. J. A., Rony, M., Majadi, N. NSTU-BDTAKA: an open dataset for Bangladeshi paper currency detection and recognition. Data in Brief 55, 2024.

[22] Sandler, M. et al. MobileNetV2: inverted residuals and linear bottlenecks. CVPR, 2018.

[23] Mutanen, J., Jaaskelainen, T., Parkkinen, J. P. S. Luminescent security properties of banknotes. IS&T PICS Conference, 2003, pp. 421-424.

[24] Nayef, N. et al. ICDAR2019 robust reading challenge on multi-lingual scene text (RRC-MLT-2019). ICDAR, 2019.

[25] Piczak, K. J. ESC: dataset for environmental sound classification. ACM Multimedia, 2015.

Resources and Links

Our code and results

SARA: glass software for the Raspberry Pi 5
github.com/aflam330/SARA

Research repository: experiments, results, thesis
github.com/aflam330/Senior_project_wearable_glass_for_blind_people_using_ocr_and_ylo

Datasets

JaalTaka (counterfeit notes)
data.mendeley.com/datasets/2m7wk5cy4c/2

BanglaTaka (note photos)
data.mendeley.com/datasets/3cv2sypkkh/1

NSTU-BDTAKA (hand-held notes)
www.sciencedirect.com/science/article/pii/S2352340924006681

Microsoft COCO
cocodataset.org

ICDAR 2019 MLT
rrc.cvc.uab.es/?ch=15

ESC-50 environmental sounds
github.com/karolpiczak/ESC-50

Open-source tools

Ultralytics YOLOv8
github.com/ultralytics/ultralytics

EasyOCR
github.com/JaidedAI/EasyOCR

Piper text-to-speech
github.com/rhasspy/piper

Vosk speech recognition
alphacephei.com/vosk/

eSpeak NG
github.com/espeak-ng/espeak-ng

WHO vision fact sheet
www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment